INCH-POUND

MIL-PRF-19500/669
16 June 2000

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON,
SURFACE MOUNT POWER RECTIFIER,
TYPES 1N6804UEG2 through 1N6810UEG2
JAN, JANTX AND JANTXV

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, hermetically sealed, surface mounted power rectifier diodes. Three levels of product assurance are provided for each device as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, (DO-214BA).

1.3 Maximum ratings.

Types	V_{RWM}	I_O T_L = +125°C 1/ 2/	I_{FSM} T_A = +100°C I_O = 750 mA dc t_p = 8.3 ms	$t_{rr\ max}$	$R_{\theta JL}$ (see figure 2) 3/	T_{STG} and T_J
	V(pk)	A dc	A(pk) max	µs	°C/W max	°C
1N6804UEG2	50	1	25	2	25	-65 to +175
1N6805UEG2	100	1	25	2	25	
1N6806UEG2	200	1	25	2	25	
1N6807UEG2	400	1	25	2	25	
1N6808UEG2	600	1	25	2	25	
1N6809UEG2	800	1	25	2	25	
1N6810UEG2	1,000	1	25	2	25	

1/ For $T_L \leq +125°C$, I_O = 1A For $T_L > +125°C$, de-rate linearly at 20 mA/°C. See figure 2.
2/ For mounting condition see figure 2.
3/ $R_{\theta JL}$ at L = PCB Mounted on 0.2 x 0.2 (5.0 x 5.0 mm) copper pad areas.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAC, P. O. Box 3990, Columbus, OH 43213-1199), by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961
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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, Building 4D (DPM-DODSSP), 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ALLIANCE (EIA)

JEDEC - JESD-22-A101 - Steady state temperature humidity bias life test

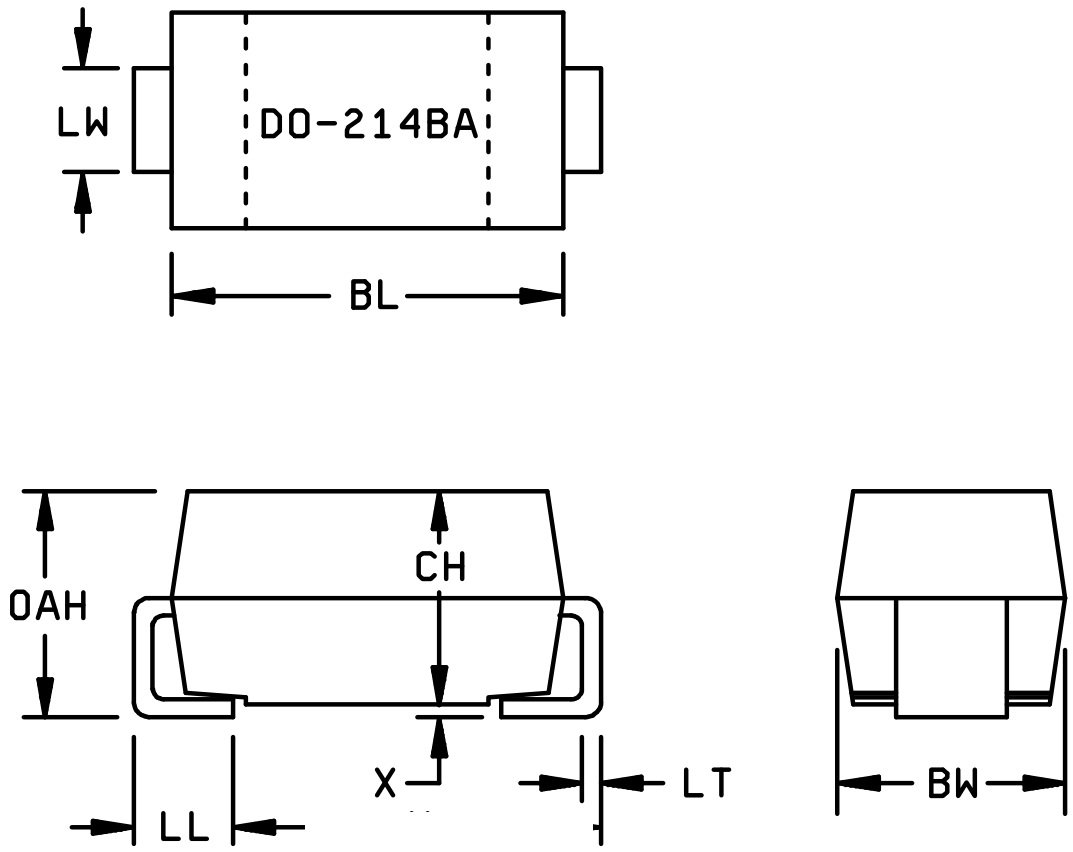
JESD-22-A102 - Accelerated moisture resistance unbiased autoclave

(Application for copies should be addressed to the Electronic Industries Alliance, 2500 Wilson Boulevard, Arlington VA 22201-3834.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.155	.189	3.95	4.80
CH	.090	.130	2.30	3.30
BW	.079	.130	2.00	3.30
LL	.029	.063	0.75	1.60
LT	.0059	.016	0.15	0.41
LW	.039	.067	1.00	1.70
OAH	.092	.138	2.35	3.5
X	.0002	.008	.005	0.20

Notes:
1. Dimensioning and tolerancing per ASME Y14.5M-1994.
2. All dimensions are in inches. Millimeters given for reference only.
3. Dimensions BL and BW do not include mold flash. Mold flash shall not exceed 0.13 mm per side

FIGURE 1. Physical dimension (DO-214BA).

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3. REQUIREMENTS

3.1 General. Individual item requirements shall be in accordance with MIL-PRF-19500 and this document herein.

3.2 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable Qualified Manufacturer's List before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

EG	Epoxy over glass
Autoclave	Accelerated moisture test performed at elevated temperature and pressure

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 and 4 herein. All devices shall be metallurgically bonded-thermally-matched-noncavity-double plug construction, utilizing a category I bond, in accordance with MIL-PRF-19500.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500, except as specified herein.

Marking information. Marking information codes are as follows:

JQ, JXQ or JVQ (J = JAN, JX = JANTX, JV = JANTXV, Q = 1N68XX where Q will be A through G to indicate the voltage rating)
Y-M (date code: Y = year, M = month)
Bar indicates cathode end

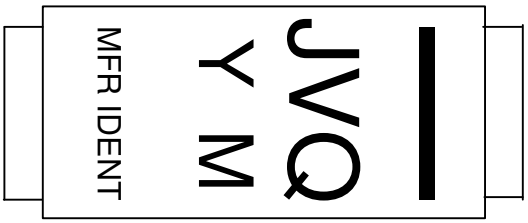
DATE CODE
1st character: designator of the manufacturing year, where Y will be "M" through "Y" to indicate the year.

2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
M	N	P	R	S	T	U	V	W	X	Y

2nd character: designator of the manufacturing month, where M will be "1" through "D" to indicate the month.

JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
1	2	3	4	5	6	7	8	9	O	N	D

A = 1N6804
B = 1N6805
C = 1N6806
D = 1N6807
E = 1N6808
F = 1N6809
G = 1N6810



3.6 Polarity. The polarity of all types shall be indicated with a visible bar to denote the cathode end.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

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3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I group A herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500. For QML see Appendix C of MIL-PRF-19500. Supplier imposed requirements shall be documented in the Quality Management Plan (see 4.2.1) and must be submitted to the Qualifying Activity for approval. Radiation characterization may be submitted in the QM plan at the option of the manufacturer.

4.3 Screening (JANTX and JANTXV levels only). For Appendix D qualified suppliers, screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable. For Appendix C qualified suppliers (QML), refer to QM plan for screening requirements.

Screen (see Appendix E, table IV of MIL-PRF-19500)	JANTX and JANTXV levels
1/	Thermal impedance (see 4.3.2)
11	I_{R1} and V_F
12	See 4.3.1
13	Subgroup 2 of table I herein: $\Delta I_{R1} \leq 100$ percent of initial reading or ± 100 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.1$ V dc.

1/ Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1038 of MIL-STD-750, condition B, T_A = room ambient as designated in 4.5 of MIL-STD-750, V_R = full rated V_{RWM} (see 1.3); f = 50-60 Hz, I_O will be adjusted to meet the requirements in paragraph 4.5.5 herein.

4.3.2 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3101 of MIL-STD-750. Maximum limit = 4.5°C/W.

4.3.2.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 (read and record value $Z_{\theta JX}$) and shall be supplied to the qualifying activity on a 500-piece sample from the qualification lot prior to qualification approval.

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4.4 Conformance inspection. For Appendix D qualified suppliers, conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. For Appendix C qualified suppliers (QML) refer to QM plan for ongoing reliability monitor requirements.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with Appendix E, table V of MIL-PRF-19500, and table I herein. $Z_{\theta JX}$ measurements in group A, subgroup 2 shall be performed in accordance with MIL-STD-750, method 3101. See 1.3 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in Appendix E, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein, delta measurements shall be as specified in table II herein. $Z_{\theta JX}$ need not to be performed. Leaded samples may be used in lieu of surface mount devices for life tests.

4.4.2.1 Group B inspection, Appendix E, table VIb (JAN, JANTX, JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1027	I_O = 1 A dc minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3). See 4.5.4 and 4.5.5
B5	3101	+25°C ≤ T_A ≤ +35°C (recorded before test is performed); $R_{\theta JL}$ (maximum) in accordance with 1.3.
B7	JESD-22-A102	Autoclave, Sample size n = 100, c = 0, 96 hrs.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in Appendix E, table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 and as specified in table II herein. Delta measurements shall be as specified in table II herein. $Z_{\theta JX}$ need not to be performed. See subgroup conditions for delta limits when applicable. Leaded samples may be used in lieu of surface mount devices for life tests.

4.4.3.1 Group C inspection, Appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Tension: Test condition A, weight = 12 pounds.
C6	1026	I_O = 1 A dc minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3). See 4.5.4 and 4.5.5.
C8	JESD-22-A101	Temperature humidity, sample size n = 100, c = 0, 1000 hrs.

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4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in Appendix E, table IX of MIL-PRF-19500.

4.4.4.1 Group E inspection, Appendix E, table IX of MIL-PRF-19500.

Subgroup	Method	Conditions
E1	1051	1000 cycles, condition G, n = 77, c = 0
E2	1037	n = 77, c = 0, 15,000 cycles
E3	2101	n = 2, c = 0
E4		N/A
E5	1001	1N6805 - 1N6808 = 8 mm Hg (100,000 ft); 1N6809, 1N6810 = 33 mm Hg (70,000 ft); $V_R = V_{RWM}$ (see 1.3); t = 1 minute (minimum); n = 15, c = 0.
E6	1020	n = 30, c = 0
E7		Parametric characterization, see figure 3, n = 25 for 3 lots

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at an ambient temperature T_A = +25°C ± 3°C.

4.5.3 Thermal resistance. Thermal resistance measurement shall be performed in accordance with method 3101 or method 4081 of MIL-STD-750. Forced moving air or draft shall not be permitted across the device during test. The maximum limit for $R_{\theta JL}$ under these test conditions shall be $R_{\theta JL(max)}$ = in accordance with 1.3.

4.5.4 Life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees nor less than 150 degrees.

4.5.5 Mounting conditions. At the option of the manufacturer, any clips or heat sink mounting configurations may be utilized provided that I_o is adjusted such that the junction temperature of each diode is maintained at least +120°C above the specified ambient temperature of +25°C.

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		Symbol	Limits		Unit	Sample plan
	Method	Conditions		Min	Max		
Subgroup 1							
Visual and mechanical inspection	2071						n = 22, c = 0
Subgroup 2							
Forward voltage	4011	I_F = 1 A dc, pulsed (see 4.5.1) t_p = 300 µs, 2% max. duty cycle	V_F	0.8	1.1	V dc	1/ 100 percent
Reverse current leakage	4016	DC method; V_R = rated V_{RWM} (see 1.3)	I_{R1}		0.5	µA dc	1/ 100 percent
Breakdown voltage 1N6804UEG2 1N6805UEG2 1N6806UEG2 1N6807UEG2 1N6808UEG2 1N6809UEG2 1N6810UEG2	4021	I_R = 50 µA dc	$V_{(BR)R}$	 50 100 200 400 600 800 1,000		V dc	1/ 100 percent
Thermal impedance	3101	See 4.3.2	$Z_{\theta JX}$		4.5	°C/W	1/ 100 percent
Subgroup 3							
High temperature operation Reverse current leakage	4016	T_A = +125°C DC method; V_R = rated V_R (see 1.3)	I_{R2}		50	µA dc	n = 45, c = 0
Subgroup 4							
Reverse recovery time	4031	Condition B1	t_{rr}		2	µs	1/ 100 percent
Subgroup 5							
Not applicable							
Subgroup 6							
Surge	4066	I_{FSM} = 25 A (pk); 10 surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 750 mA dc; V_R = rated V_{RWM} (see 1.3); T_A = +100°C					n = 22, c = 0
End points		Group A, subgroup 2					
Subgroup 7							
Not applicable							

1/ 100 percent not required if performed 100 percent in screen 13. Sample plan will then be n = 116, c = 0.

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TABLE II. Groups B, and C delta measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits	
		Method	Conditions		Min	Max
1.	Reverse current leakage change	4016	DC method	ΔI_{R1} 3/		for JANTX and JANTXV, ≤ 250 nA dc or 100 percent, whichever is greater.

1/ The electrical measurements for Appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table II herein, step 1.
   b. Subgroup 6, see table II herein, step 1.
2/ The electrical measurements for Appendix E, table VII of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table II herein, step 1.
   b. Subgroup 6, see table II herein, step 1 (JANTX and JANTXV).
3/ Devices which exceed the group A limits for this test shall not be accepted.

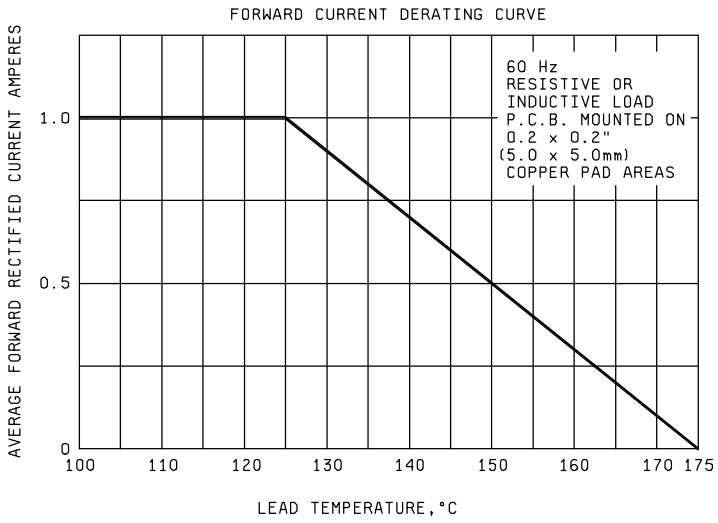
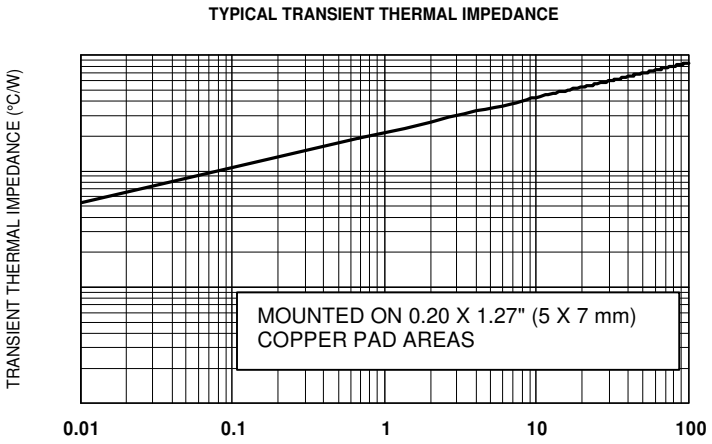


FIGURE 2. Power derating curves maximum power in watts versus lead temperature.

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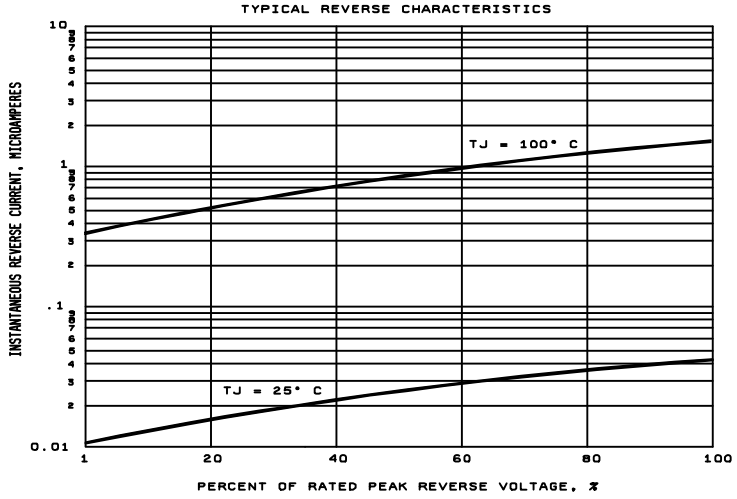
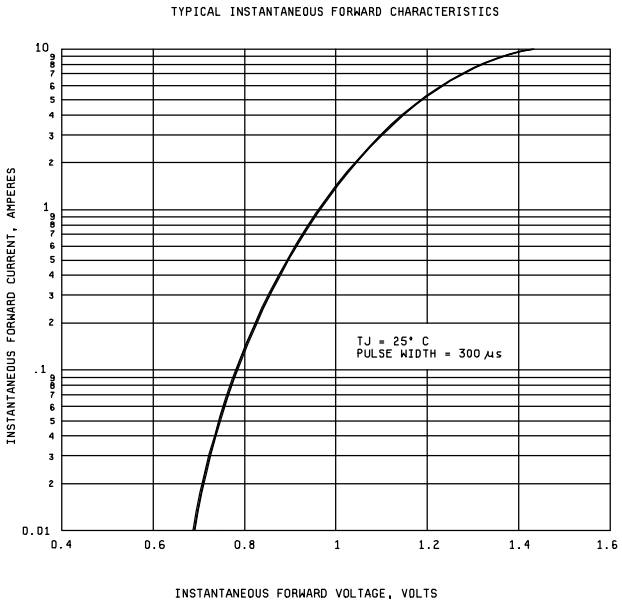


FIGURE 3. Parametric characterization.

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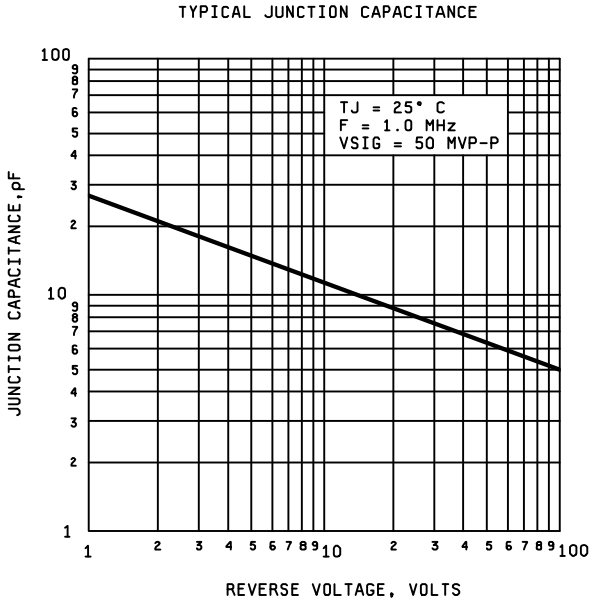


FIGURE 3. Parametric characterization, continued.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Departments or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

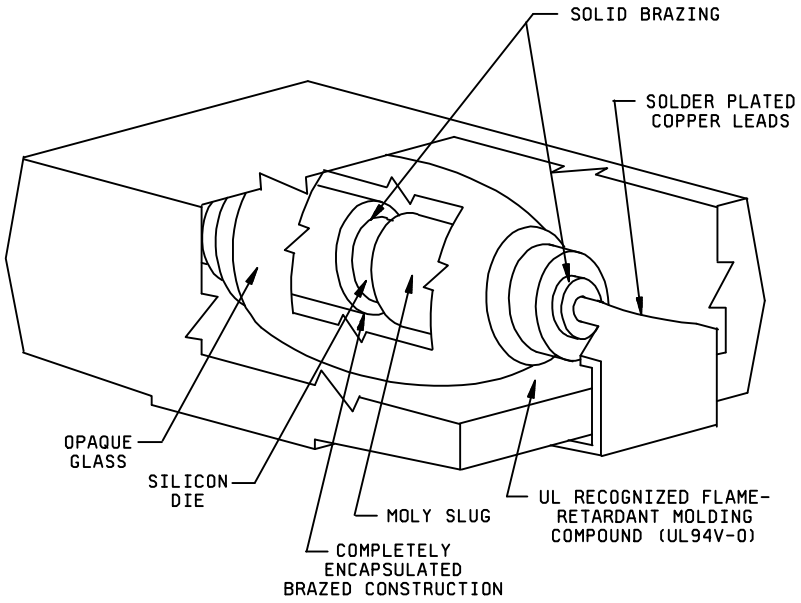


FIGURE 4. Sample construction cross section.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Issue of DoDISS to be cited in the solicitation (see 2.2.1).

b. The lead finish as specified (see 3.4.1).

c. Type designation and quality assurance level.

d. Packaging requirements (see 5.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43213-1199.

6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer and user Part or Identifying Number (PIN). This information in no way implies that the vendor similar part numbers are substitutable for the military PIN.

Preferred types military PIN	Vendor similar part number
1N6804UEG2	GF1A
1N6805UEG2	GF1B
1N6806UEG2	GF1D
1N6807UEG2	GF1G
1N6808UEG2	GF1J
1N6809UEG2	GF1K
1N6810UEG2	GF1M

6.4.1 Substitution data. The following MIL-PRF-19500/669 devices are not intended to replace the MIL-PRF-19500/427 US & UL types, but are offered as an option where strain relief is of concern.

MIL-PRF-19500/427 devices	MIL-PRF-19500/669 devices
1N5614US,UL	1N6806UEG2
1N5616US,UL	1N6807UEG2
1N5618US,UL	1N6808UEG2
1N5620US,UL	1N6809UEG2
1N5622US,UL	1N6810UEG2

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2186)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/669	2. DOCUMENT DATE 16 June 2000

3. DOCUMENT TITLE: SEMICONDUCTOR DEVICE, DIODE, SILICON, SURFACE MOUNT POWER RECTIFIER, TYPES 1N6804UEG2 through 1N6810UEG2 JAN, JANTX AND JANTXV

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (PLEASE PRINT)	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan_barone@dscc.dla.mil
c. ADDRESS *(Include Zip Code)* DSCC-VAC 3900 E. Broad Street Columbus, Ohio 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703)767-6888 DSN 427-6888

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